

Week 6, October 9, 2024

Main topics

- Choice of Regularization Parameter.
- Iterative Methods for Large-Scale inverse Problems.

Material

From last week: P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, Chapter 5. Supplement by Kirsch, section 2.5.

New material this week: P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, Chapter 6. Supplement by Kirsch, section 2.6.

Program

13:00-13:30 Choice of Regularization Parameter – short lecture.

13:30-15:45 Exercises (see below).

14:45-16:00 Break

16:00-17:00 Lecture: Iterative Methods for Large-Scale Problems.

Exercises

1. Spend 5-10 minutes giving a bit of feedback to the course via <https://evaluating.dtu.dk/>.

2. **Proof of theorem**

Last week we considered the data misfit function $F_1(\alpha) = \|Kx_\alpha - y^\delta\|$ with $y^\delta = Kx^* + n$, $\|n\| \leq \delta$ and $x_\alpha = R_\alpha y^\delta$ the Tikhonov solution.

We saw the theorem about a unique α^* with $F_1(\alpha^*) = \delta$; picking the regularization parameter this way is the parameter choice rule suggested by the Morozov discrepancy principle.

Let's look at the discrete case $K = A \in \mathbb{R}^{n \times m}$ and carry out the main steps of the proof. We will write $P^\perp$ for the orthogonal projection onto $(\text{Col } A)^\perp = (\text{Ran } K)^\perp$ and assume that $\|P^\perp y^\delta\| < \delta$ and $\|y^\delta\| > \delta$. Is this reasonable?

- (a) Show that $\lim_{\alpha \rightarrow 0} F_1(\alpha) = \|P^\perp y^\delta\| < \delta$.
- (b) Show that F_1 is continuous and strictly increasing.

(c) Show that $\lim_{\alpha \rightarrow \infty} F_1(\alpha) = \|y^\delta\| > \delta$.

Illustrate the above findings in a plot of the function $F_1(\alpha)$ for $\alpha \in (0, \infty)$.

Hints:

(a) Show and use that

$$Kx_\alpha - y^\delta = \sum_{\mu_j > 0} \left(\frac{\mu_j^2}{\mu_j^2 + \alpha} - 1 \right) (y^\delta, y_j) y_j + P^\perp y^\delta.$$

(b) We will do this in class.

(c) Establish the inequalities

$$\alpha \|x_\alpha\| \leq J_\alpha(x_\alpha) \leq J_\alpha(0) = \|y^\delta\|$$

to draw the conclusion.

What changes in the above argument, when $A = K$ is a compact operator between Hilbert spaces?

3. **Computational investigation of $F_1(\alpha)$.** Take the Abel problem with $x(1) = 1, y(t) = 2\sqrt{t}$. Discretize with 128 points, add Gaussian random noise with $\delta = 5\%$. For α from 10^{-4} to 1 (step .2), compute $F_1(\alpha)$ and make a loglog plot of $F_1(\alpha)$ against α . Indicate in the plot the noise level. Experiment with other levels of noise.

4. The Discrepancy Principle

The purpose of this exercise is to illustrate the sensitivity of the discrepancy principle to variations of the estimate of the error norm $\|e\|_2$.

First generate the deblurring model problem for $n = 100$. This problem has the exact solution pair

$$x(s) = \phi(s), \quad y(t) = (6 - |t|)(1 + .5 \cos(t\pi/3)) + 9/(2\pi) \sin(|t|\pi/3).$$

Make discrete approximations of the functions on a uniform mesh on $(-6, 6)$. Add Gaussian noise with standard deviation $\eta = 0.01$ to the right-hand side: $y^\delta = y + e$, $e = \text{eta} * \text{randn}(n, 1)$.

Use the discrepancy principle to compute the Tikhonov solution.

Take the safety factor $\tau = 1$. Try to use both the norm estimate $\sqrt{n} \eta$ and the actual 2-norm $\delta = \|e\|_2$ of the perturbation vector e . Is there a significant difference in the results? NB: try with different seeds for `randn`.

Now replace $\|e\|_2$ with values that are too small and too large; this simulates a situation where only a rough estimate is available. For example, scale δ with factors 0.8, 0.9, 0.95, 1.3, and 10. How sensitive is the regularized solution to such under/overestimates?

5. The L-Curve Method

This exercise illustrates the use of the L-curve method for choosing the regularization parameter α in Tikhonov regularization. We use the `phillips` test problem with $n = 64$ and standard deviation 0.01 of the noise.

Plot the L-curve.

In your plot of the L-curve you can add the point that corresponds to the optimal reg. parameter α_{opt} , i.e., the one that minimizes the error $\|x^* - x_\alpha\|_2$. How close are these points to the ones found by the parameter-choice methods?